

IDS-101-09: College Colloquium: Thinking Machines

Calvin Deutschbein (they) and Addie Gage (she)

Willamette University, Fall 2025

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Office Hours: MWF 11:30-12:00 & Discord TTh
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Web: cd-public.github.io/
Lecture: MWF 12:00-1:00
Chase Hour: Th 12:00-1:00

Course Description

College Colloquium

Topical seminars designed to pursue significant issues and questions of special interest to instructors and students. Seminars invite students into the intellectual life of the university, model rigorous engagement, and help them develop qualities of good scholarship – effective writing, careful reading, critical thinking, and cogent argumentation. Seminars do not count toward majors or minors. Required for all entering firstyear students.

Thinking Machines

Artificial intelligence, machine learning, big data, and large scale computing have captured the collective imaginations of millions people across a range of technical expertise. Yet each of these buzzwords is rooted in a central promise: that machines may think.

Guided by student inquiry, we will learn how to construct thinking - or the appearance thereof - from basic computation. Using the shared language of computer code, we use programming as a way to bridge the gap between thinking humans and thinking machines. With this shared experience, we will be able to reflect more thoughtfully on the implications of thinking machines on the self, culture, and society.

About Us

Calvin Deutschbein is a fifth-year professor of computer science and colloquia instructor.

Addie Gage is a fourth-year student at Willamette studying Computer Science and Data Science in the MS Data Science program, and a STEM fellow.

Required Materials

Required materials for a given class will be available on the [course webpage](#). All course materials will be made available at no cost to the student.

Accessibility

I will make every effort to ensure all coursework and materials are accessible to all students, including working with on-campus specialists. However, there is always room for improvement. I always appreciate hearing from students about how I can make the course more accessible, so please reach out if there is something I can be doing better!

Course Objectives

Per the college, we learn and practice techniques to:

- Critically examine information and/or texts (written, oral, artistic, or quantitative) by identifying central ideas or arguments, making inferences, questioning underlying assumptions, and assessing evidence.
- Learn to contribute to a constructive classroom climate through participation in thoughtful, informed, and responsive discussion, effective speaking, active listening, and the development of an iterative group process of critical analysis and interpretation.
- Effectively formulate ideas and arguments, develop them through an iterative process, and express them clearly and persuasively via linguistic, artistic, and/or quantitative modes of communication.

Additionally, I will attempt to construct the class in a manner that these techniques specifically create an inclusive academic environment.

Course Structure

The course will be composed of lecture, demonstration, and labs on scientific computing, and out-of-class writing of both technical and non-technical nature.

Students will also read and respond to each other's writing within class.

Class Structure

Classes are scheduled for Monday, Wednesday, and Friday at 12:00 noon. There is a special supporting CHASE session on Thursdays at 12:00 noon. The lecture schedule is on the [course webpage](#).

Papers/Projects

There will be both a midterm paper/project and final paper/project for this course. Though we term them papers, they can more properly be considered scientific publications. The midterm will be a source code repository on GitHub with both Python code and Markdown documentation. The final will be a GitHub Pages website created using the Quarto platform for scientific publication. These tools will be taught during the course.

The midterm will be due 10 October at 12:00 PM. The final paper will be due 14 November at 12:00 PM.

Feedback and Grading

Exercises

- For an A (100%)
 - Code that meets requirements.
 - On time.
- For a B (90%)
 - Code that runs without error.
 - On time.
- Erroneous code is an F (60%).
- Late work is a zero (0%).

Papers

Graded on **AP Scale** as an argument essay in support (or against) scientific computing. If I take more than one week to return grades, I provide “grading amnesty” of up to a letter grade.

Weighting

There will be 9 exercises of equal weight, and midterm weighted as 3 exercises and a final weighted as 6 exercises. I will drop the three lowest (which could be up to half the final or the entire midterm) and take the average of the top 15 scores.

Students should expect to earn to perfect scores on exercises unless skipping class, which essentially amounts to an attendance incentive of 60% of the final grade.

College Policies

The following material is adapted from “Information for Syllabus” recommended language on syllabus preparation provided to instructors in the College of Arts & Sciences.

Academic Integrity

Students of Willamette University are members of a community that values excellence and integrity in every aspect of life. As such, we expect all community members to live up to the highest standards of personal, ethical, and moral conduct. Students are expected not to engage in any type of academic or intellectually dishonest practice and encouraged to display honesty, trust, fairness, respect, and responsibility in all they do. Plagiarism and cheating are especially offensive to the integrity of courses in which they occur and against the College community as a whole. These acts involve intellectual dishonesty, deception, and fraud, which inhibit the honest exchange of ideas. Plagiarism and cheating may be grounds for failure in the course and/or dismissal from the College. <http://willamette.edu/cla/catalog/policies/plagiarism-cheating.php>

Commitment to Positive Sexual Ethics

Willamette is a community committed to fostering safe, productive learning environments, and we value ethical sexual behaviors and standards. Title IX and our school policy prohibit discrimination on the basis of sex, which regards sexual misconduct — including discrimination, harassment, domestic and dating violence, sexual assault, and stalking. We understand that sexual violence can undermine students’ academic success, and we encourage affected students to talk to someone about their experiences and get the support they need.

Please be aware that as a mandatory reporter I am required to report any instances you disclose to Willamette's Title IX Coordinator.

If you would rather share information with a confidential employee who does not have this responsibility, please contact our confidential advocate at confidential-advocate@willamette.edu. Confidential support also can be found with SARAs and at the GRAC (503-851-4245); and at WUTalk - a 24-hour telephone crisis counseling support line (503-375-5353). If you are in immediate danger, you may reach campus safety at 503-370-6911.

DACA/Undocumented Student Advocate

Willamette is committed to supporting our DACA/Undocumented students in a variety of ways. This year, Tori Ruiz is the contact person for all DACA/undocumented students can provide those students with a number of external and internal resources that are available. Her contact information is email: truiz@willamette.edu, Office: 3rd Floor UC, Phone: 503-370-6447.

Diversity and Disability Statement

Willamette University values diversity and inclusion; we are committed to a climate of mutual respect and full participation. My goal is to create a learning environment that is usable, equitable, inclusive and welcoming. If there are aspects of the instruction or design of this course that result in barriers to your inclusion or accurate assessment or achievement, please notify me as soon as possible. Students with disabilities are also encouraged to contact the Accessible Education Services office in Smullin 155 at 503-370-6737 or Accessible-info@willamette.edu to discuss a range of options to removing barriers in the course, including accommodations.

Religious Practice

Willamette University recognizes the value of religious practice and strives to accommodate students' commitment to their religious traditions whenever possible. Please let me know within the first two weeks of the semester if a conflict between holy days or other religious practice and full participation in the course is anticipated. I will do my best to work with you to determine a reasonable accommodation.

As an instructor, I will exercise my discretion to offer accommodations for conflicts after the first two weeks of the semester. You may always reach out to me, including retroactively, though the quality of the accommodation I am able to offer may improve given advanced warning!

SOAR Center Offerings: Food, Clothing, and School Materials

The Students Organizing for Access to Resources (SOAR) Center strives to create equitable access to food, professional clothing, commencement regalia, and scholarly resources for WU and Willamette Academy students. The SOAR Center is located on the Putnam University Center's third floor (in the former Women's Resource Center and across from the Harrison Conference Room). The space houses the Bearcat Pantry, Clothing Share, and First-Generation Book Drive and is maintained by committed students and staff and faculty advisers.

Trans Inclusion and Gender Justice

I am always appreciative of the opportunity to address you by your affirming name or pronoun. Please advise me of the most affirming way to address you at any time so that I may do so.

If I ever misgender you in any way, I would greatly appreciate that you let me know, in whatever manner makes you comfortable, so that I can correct that error and endeavour to repair any harm.

Mental Health

As a student, you may experience a range of challenges that can interfere with learning, such as strained relationships, increased anxiety, substance use, feeling down, difficulty concentrating and/or lack of motivation. These mental health concerns or stressful events may diminish your academic performance and/or reduce your ability to participate in daily activities. Willamette services are available and treatment does work. If you think you need help, please contact Bishop Health as soon as possible at <http://willamette.edu/offices/counseling/>. Crisis counseling is available 24/7 at WUTalk: 503-375-5353 and Campus Safety is available at 503-370-6911. Emergency resources are also available from the Psychiatric Crisis Center at 503-585-4949 and the National Suicide Prevention Lifeline at 1-800-273-8255.